AUTNES Automatic Content Analysis of the Media Coverage 2024

Documentation V1.0

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Introduction

This Documentation Report accompanies and describes the data of the AUTNES Automatic Content Analysis of Media Coverage of the Austrian National Election held on 29 September 2024. The body elected was the National Parliament (Nationalrat).

The Documentation Report is structured in six sections. The first s ection o utlines the conditions of use of the data. The second section, the study description, provides an overview of the study. The third section presents the study design, i.e., the media sample and the sampling procedure. The fourth section provides information about the coding procedure and validation. The last section provides an overview of of all variables in the analyzed media outlets.

Acknowledgement of the data

Data users are required to acknowledge the data and the accompanying release document. Please refer to the AUSSDA data catalogue (https://data.aussda.at, doi: 10.11587/HOEXGX) for a recommendation on how to cite these data and the documentation.

1 Conditions of Use

1.1 Restrictions

The data are available under the terms listed at the AUSSDA Dataverse.

1.2 Confidentiality

AUTNES, the Principal Investigators and the funding institution bear no responsibility for the use of the data, or for interpretations or inferences based on their use, neither do they accept liability for indirect, consequential or incidental damages or losses arising from the use of the data.

1.3 Deposit Requirement

To facilitate exchanges within the scientific community and to provide funding agencies with essential information about the use of archival resources, users of the AUTNES data are requested to notify the AUTNES team of all forms of publications referring to AUTNES data at jana.bernhard@univie.ac.at.

2 Study description

2.1 Title and Edition

AUTNES Automatic Content Analysis of the Media Coverage 2024

2.2 Research Team

Jana Bernhard-Harrer, Marvin Stecker, Jakob-Moritz Eberl, Hajo G. Boomgaarden (Media Side)

Sylvia Kritzinger, University of Vienna (Demand Side)

Laurenz Ennser-Jedenastik, University of Vienna (Supply Side)

2.3 Acknowledgements

This data collection has been enabled though the APA-UNIVIE Data Project.

2.4 File Overview

File Name	Format	Variables	Observations
`10906_da_en_v1_0.dta`	STATA	52	50,042 paragraphs
`10906_da_en_v1_0.csv`	CSV	52	50,042 paragraphs
`10906_da_en_v1_0.zsav`	SPSS	52	50,042 paragraphs
`10906_da_en_v1_0.feather`	FEATHER	52	50,042 paragraphs

2.5 Topics /Keywords

Election coverage, Political actors, Mass media coverage, Elections, Election campaigns, Political parties, Political representation, Political issues, Content Analysis

3 Study design

The data set is based on the automatic analysis of all political news reports published in Austrian media outlets (print, online, TV) leading up to the Austrian National Election 2024. We define a news report as political when the report mentions a central political actor. Political actors in that sense are all members of the government, all lead candidates, and the parties themselves that were represented in the parliament before or after the elections.

Central political actors are

- designations such as government (coalition, governing parties) and opposition (opposition parties) on the national level
- members of the government before the election, on national, state and EU level
- ministries on the national level
- the Austrian President
- Austrian political parties eligible on the national level, including demographically defined sub-organizations on the national level (e.g. for women)
- political candidates running for parliament or office in the election

If any of these actors is mentioned in a news item, it was considered political news. The period of analysis ranges from 25 August to 29 September 2024. The media data can be synced with AUTNES demand side data, specifically with survey wave 22 of the AUTNES On-line Panel Study (see Online Panel Study 2017-2024, doi:10.11587/HNUFCC, and Partheymüller et al.[1] for more information on and documentation for the survey data).

3.1 Media Sample

The media sample was selected based on the type and circulation to reflect the Austrian national media landscape. All media reports that mentioned a relevant political actor were downloaded from the APA (Austrian Press Agency) database, via the APA-API Data Project. The data includes reports from 16 media outlets, covering print media, TV, and online media. For an overview see Table Table 2.

Table 2: Overview of media outlets and number of articles

Media Outlet	Type	N
Der Standard	Quality Print	231
Die Presse	Quality Print	241
Heute	Tabloid Print	337
Kleine Zeitung	Mid-Market Print	639
Kronen Zeitung	Tabloid Print	1214
Kurier	Mid-Market Print	535
Salzburger Nachrichten	Quality Print	346
ZiB	Public Broadcaster	102
derstandard.at	Quality Online	302
diepresse.com	Quality Online	244
heute.at	Tabloid Online	265
kleine.at	Mid-Market Online	452
krone.at	Tabloid Online	526
kurier.at	Mid-Market Online	280

[1] Partheymüller, J., Kritzinger, S. & Wagner, M. The AUTNES Online Panel Study 2017–2024: A Dataset of Austrian Voter Attitudes and Behavior. Sci Data 12, 1487 (2025). https://doi.org/10.1038/s41597-025-05848-2 Dataset availabe at: https://doi.org/10.11587/HNUFCC

Table 2: Overview of media outlets and number of articles

Media Outlet	Type	N
oe24.at	Tabloid Online	241
Österreich	Tabloid Print	399

3.2 Sample of News Items

To collect the data, articles for print and online newspapers, as well as transcripts for the main TV news were downloaded from the database of the Austria Presse Agentur (APA, www.apa-defacto.at), using the following search string containing the names of all Austrian political parties participating in the National Election of 2024, the names of the chancellor, all ministers, as well as the national parliament as an institution.

Search String

```
nationalrat* OR NEAR.3(österreichisch*, bundesregierung) OR övp* OR
neue* volkspartei OR volkspartei OR spö* OR sozialdemokraten OR
sozialdemokratische partei österreichs OR fpö* OR freiheitlich* OR
freiheitliche partei österreichs OR neos* OR türkis-blau OR
türkis-grün OR türkis-rot OR schwarz-blau OR schwarz-grün OR
schwarz-rot OR blau-schwarz OR blau-rot OR blau-grün OR rot-grün-pink
OR rot-grün OR rot-schwarz OR hans peter doskozil OR hans-peter
doskozil OR peter doskozil OR peter kaiser OR johanna mikl-leitner OR
thomas stelzer OR wilfried hauslauer OR christopher drexler OR anton
mattle OR markus wallner OR michael ludwig OR rot-grün OR
schwarz-grün OR (("die grüne*" OR "der grüne*" OR den grünen) NOT
(NEAR.2(grün*, rasen) OR NEAR.2(grün*, wiese*))) OR ökopartei OR
grünpolitiker* OR NEAR.4(grün*, politiker*) OR tiroler grüne* OR
grüne* tirol OR wiener grüne* OR grüne* wien OR grüne* burgenland OR
burgenländ* grüne* OR grüne* kärnten OR kärntner* grüne* OR grüne*
salzburg OR salzburger* grüne* OR grüne* oberösterreich OR
oberösterreichische* grüne* OR grüne* steiermark OR steirische*
grüne* OR grüne* vorarlberg OR vorarlberger grüne* OR grüne*
niederösterreich* OR niederösterreichische* grüne* OR kpö OR
kommunistische partei österreich OR kommunist* OR NEAR.3(wandel,
partei) OR keine von denen OR NEAR.5(wandel, mulla) OR NEAR.4(wlazny,
Bier) OR bierpartei OR bier partei OR lmp OR liste madeleine petrovic
OR nehammer* OR kanzler* nehammer OR bundeskanzler* nehammer OR
NEAR.5(spitzenkandi*, nehammer) OR karl nehammer OR babler* OR
andreas babler OR NEAR.5(spitzenkandi*, babler) OR kickl* OR herbert
kickl OR NEAR.5(spitzenkandi*, kickl) OR werner kogler OR
NEAR.5(spitzenkandi*, kogler) OR NEAR.5(kogler, grün*) OR
```

```
NEAR.5(spitzenkandi*, kogler) OR vizekanzler* kogler OR
meinl-reisinger OR NEAR.5(beate, meinl) OR NEAR.5(beate, reisinger)
OR NEAR.5(spitzenkandi*, meinl, reisinger) OR NEAR.5(spitzenkandi*,
schweiger) OR tobias schweiger OR wlazny* OR NEAR.5(spitzenkandi*,
wlazny) OR dominik wlazny OR petrovic* OR NEAR.5(spitzenkandi*,
petrovic) OR madeleine petrovic OR mulla* OR NEAR.5(spitzenkandi*,
mulla) OR fayad mulla OR minister* brunner OR finanz* brunner OR
bundsminister* brunner OR minister* edtstadler OR verfassung*
edtstadler OR bundeskanzleramt* edtstadler OR eu edtstadler OR
bundesminister* edtstadler OR minister* gewessler OR klima* gewessler
OR umwelt* gewessler OR energie* edtstadler OR mobilität* gewessler
OR innovation* gewessler OR technologie* gewessler OR bundesminister*
gewessler OR minister* karner OR innenminister* karner OR
bundesminister* karner OR minister* kocher OR arbeit* kocher OR
wirtschaft* kocher bundesminister* kocher OR minister* polaschek OR
bildung* polaschek OR wissenschaft* polaschek OR forschung* polaschek
ORbundesminister* polaschek OR minister* raab OR frauen* raab OR
familie* raab OR integration* raab OR medien* bundeskanzleramt* raab
OR bundesminister* raab OR minister* rauch OR soziales* rauch OR
gesundheit* rauch OR pflege* rauch OR konsumentenschutz* rauch OR
bundesminister* rauch OR minister* schallenberg OR international*
schallenberg OR europa* schallenberg OR außen* schallenberg OR
bundesminister* schallenberg OR minister* tanner OR verteidigung*
tanner OR bundsminister* tanner OR minister* totschnig OR
landwirtschaft* totschnig OR forstwirtschaft* totschnig OR regionen*
totschnig OR wasserwirtschaft* totschnig OR bundsminister* totschnig
OR minister* zadić OR justiz* zadić OR bundsminister* zadić OR "van
der bellen" OR (bundespräsident* NOT Steinmeier)
```

Articles were first deduplicated based on text, date, and source. In a second, more thorough step, articles were deduplicated based on textual similarity. This process involved calculating the cosine similarity between the average word embeddings of the articles' text using pre-trained word vectors. Articles from the same source with publication dates within ± 1 day were compared, and if their similarity score exceeded 0.90, one of the articles was removed to ensure only unique content remained.

The data collection and the subsequent cleaning resulted in a final data set consisting of *6354* articles or *50042* paragraphs. The subsequent analysis has been done on the paragraph level.

4 Coding Procedure

4.1 Overview

We used Lucene syntax search strings to measure Issues and translated to Lucene search strings to Regex for Actors. Wherever possible, we built on already developed search strings from the AUTNES 2017, and 2019 Automatic Content Analysis. These were available for all parties included in our data set. Regarding Lead Candidates, we had to create new search strings for Andreas Babler (SPÖ), Karl Nehammer (ÖVP) and Herbert Kickl (FPÖ). As all of these three have distinct names, the searchstrings were adapted based on the candidate search strings from prior studies.

As in previous AUTNES media analyses (2017, and 2019), we applied the *583* issue search strings that were originally developed for the project. These search strings were then aggregated into overarching topics, resulting in *20* issues. We reused most of the AUTNES issue search strings since 2013. However, in order to remain consistent with the AUTNES Panel Survey, we also included four new issues: *COVID*, *Pensions*, *Inflation*, and *Rent*. Please see *IssueSearchstrings.csv* for an overview of all issue searchstrings and *ActorSearchstrings.csv* for all actor searchstrings. For sentiment analysis, we made use of two already existing dictionaries on sentiments in German language (Rauh and SentiWS).

4.2 Validation

For the validation of the newly developed issue search strings, we conducted a manual coding process to ensure their accuracy. While most issue search strings were reused from previous AUTNES studies (2017, and 2019) and had already been validated, the new search strings for Pensions, Inflation, and Housing required additional validation. To this end, we provided student assistants with a sample of 100 newspaper articles per issue, which had been automatically assigned to the respective topics using the search strings. The assistants were instructed to manually verify whether the articles indeed addressed the intended issue, even if it was not the central focus of the article. They coded each article as "Yes" (if the issue was mentioned), "No" (if the issue was not mentioned), or "Unclear" (if they were unsure). Articles marked as "Unclear" were reviewed by the project team. This validation process ensured the reliability of the new issue search strings before their application in the analysis. The results of the validation process are summarized below, including precision, recall, and F1 scores for each issue:

Issue	Precision	Recall	F1 Score
COVID	0.83	1.00	0.91
Rent	0.50	1.00	0.67
Inflation	0.46	1.00	0.63
Pensions	0.37	1.00	0.54

These metrics highlight the trade-offs between precision and recall for the newly developed search strings. While recall was consistently high across all issues, precision varied, indicating that some search strings captured a broader range of articles, including those not directly relevant to the intended issue. The F1 score, which balances precision and recall, provides an overall measure of the search string performance.

It is important to note that the validation process for Rent, Inflation, and Pensions involved comparisons against closely related topics, which contributed to the lower precision scores. Specifically, for Pensions, all false positives were articles related to the broader topic of Wohlfahrtsstaat (welfare state). Similarly, for Inflation, all false positives were articles categorized under the topic of Economy. For Rent, the false positives also stemmed from articles about Wohlfahrtsstaat. This overlap with closely related topics reflects the challenge of distinguishing between nuanced issue boundaries in automated content analysis.

For validation of the other issues, we defer to the results and documentation of previous years. The empirical setup, such as the selection of newspapers, are similar enough that we trust these search strings to still identify relevant text passages in the articles.

5 Dataset

This chapter provides documentation on the variables included in the dataset. The dataset is structured at the paragraph level. Each paragraph is uniquely identified by the variable v0, which combines the article ID and the paragraph number (e.g., articleID_paragraphNumber). The variable v0_article holds the article ID, which can be used to aggregate the data at the article level. To aggregate the dataset on an article level you can run the following code in R:

The following R code demonstrates how to aggregate the dataset at the article level based on the specified rules:

```
# Aggregation logic
aggregated_data <- data %>%
    group_by(v0_article) %>%
    summarise(
        v0 = list(v0), # Store all v0 codes in a list
        v1 = first(v1), # v1 is the same for all
        v2 = first(v2), # v2 is the same for all
        across(starts_with("v3"), first),
        # All variables starting with v3 are the same
        v4 = first(v4), # v4 is the same
        v5 = first(v5), # v5 is the same
        v6 = first(v6), # v6 is the same
        v7 = sum(v7), # v7 is the sum
        across(starts_with("v9"), ~ list(.x)),
        # All variables starting with v9 are stored in a list
        across(starts_with("v10"), ~ list(.x)),
```

```
        # All variables starting with v10 are stored in a list
        across(starts_with("v11"), ~ list(.x)),
        # All variables starting with v11 are stored in a list
        across(starts_with("v19"), sum),
        # All variables starting with v19 are aggregated (summed)
        across(starts_with("v20"), sum)
        # All variables starting with v20 are aggregated (summed)
    )
```

5.1 Metadata

Metadata were derived from the source texts automatically via the APA API, and when necessary were manually corrected or recoded.

- Paragraph ID (v0): article id plus a numeric value that denotes the position of the paragraph within the article.
- Article ID (v0_article): randomly generated ID Number between 1000 and 100000.
- Headline (v1): Headline of the news report, stored as a string variable.
- Subheader (v2): Subheader of the news report, stored as a string variable.
- Medium Outlet (v3_v1-v3_v3; v3_l1 - v3_l3): Reported on three different levels. Variables with v3_v1 to v3_v3 are numeric. Varibales v3_l1 to v3_l3 are string labels. Level 1 (v3_l1) reports the type of media (TV, print, online, radio), level 2 (v3_l2) aggregates media outlets to subclasses (e.g., Tabloids) and level 3 (v3_l3) reports the specific media outlet in which the article was published. The table below provides an overview of the included media outlets and their corresponding levels:

v3_v1	v3_l1	v3_v2	v3_l2	v3_v3	v3_l3
1000	TV	1100	Public Broadcaster	1102	ZiB
2000	Print	2100	Quality Print	2101	Der Standard
3000	Online	3100	Quality Online	3101	derstandard.at
2000	Print	2100	Quality Print	2102	Die Presse
3000	Online	3100	Quality Online	3102	diepresse.com
2000	Print	2100	Quality Print	2103	Salzburger Nachrichten
2000	Print	2200	Tabloid Print	2201	Kronen Zeitung
3000	Online	3200	Tabloid Online	3201	krone.at
2000	Print	2200	Tabloid Print	2202	Österreich
3000	Online	3200	Tabloid Online	3202	oe24.at
2000	Print	2200	Tabloid Print	2203	Heute
3000	Online	3200	Tabloid Online	3203	heute.at
2000	Print	2300	Mid-Market Print	2301	Kurier
3000	Online	3300	Mid-Market Online	3301	kurier.at
2000	Print	2300	Mid-Market Print	2302	Kleine Zeitung
3000	Online	3300	Mid-Market Online	3302	kleine.at

- Date (v4): The date on which the article was published.
- Page (v5): The page on which the article was published. Only reported for articles in print media (when available).
- Section (v6): Automatically coded variable on the section in which the article was published. A section was only assigned when possible. Sections are only available for print and online media. See Table below for the harmonzied Section Names.

Harmonized	Category (APA label)
1	Front page / Titelseite
2	Politics / Domestic
3	Thematic focus / Issues
4	Local / Regional / Chronik
5	International / Foreign
6	Feuilleton
7	Economy / Business
8	Culture / Arts
9	Media / Communication
10	Society
11	Science / Innovation
12	Sports
14	Opinion / Debate
15	Television / Programs
16	Lifestyle / Leisure
17	Reports / Features
33	Supplements / Services
99	Miscellaneous / Inserts

- Word count (v7): Length of main text of articles, coded automatically.

5.2 Parties

The appearance in news reports of the main parties in the 2024 national election (SPÖ, ÖVP, FPÖ, Greens, NEOS) was coded with the same search strings as in the 2019 National Election. All party variables have the value 1 when the given party is mentioned in the news report, and 0 if it is not mentioned.

- FPÖ (v9a)

```
fpo freiheitlich* fp "partei freiheitlich*"~5
```

[Precision: 0.94 | Recall: 0.92]

- ÖVP (v9b)

```
ovp vp volkspartei "liste kurz"~5 "neue* volkspartei" "die schwarzen"
"den schwarzen" "die turkisen" "den turkisen"
```

[Precision: 0.94 | Recall: 0.93]

- SPÖ (v9c)

```
spo sp "partei sozialdemo*"~5 sozialdemokrat* "die roten"
"den roten" sozi sozialist*
```

[Precision: 0.94 | Recall: 0.94]

- Grüne (v9d)

```
("grune partei" "die grunen" "die grune" "der grunen" "der grune"
"die grune partei" "grune alternative" okopartei grunpolitiker*
"grun politiker*" "den grunen"

"tiroler grune*" "wiener grune*" "grune* tirol" "grune* wien"
"grune* burgenland" "burgenland* grune*"
"grune* karnten" "karntner grune*"
"grune* salzburg" "salzburger grune*"
"grune* oberosterreich" "oberosterreichische* grune*"
"grune* steiermark" "steirische* grune*"
"grune* vorarlberg" "vorarlberger grune*"
"grune* niederosterreich*" "niederosterreichische* grune*"

"grune* spitzenkandidat*"~5 "grune* vorstandssprecher*"~5
"grune* landtagsabgeordnet*"~5 "grune* chef*"~5 "grune* vorsitz*"~5
"grune* felipe"~5 "grune* kogler"~5 "grune* sprecher*"~5
"grune* bundesvorstand*"~5 "grune* bundessprecher*"~5
"grune* landesvorstand*"~5 "grune* parlament*"~5 "grune* regierung*"~5
"grune* opposition*"~5 "grune* partei*"~5 "grune* bundnis*"~5
"grune* fraktion*"~5 "grune* klub*"~5 "grune* vize*"~5
"grune* burgermeister"~5 "grune* stadtrat*"~5 "grune* landesrat*"~5
"grune* bereichssprecher*"~5 "grune* referent*"~5
"grune* landespart*"~5 "grune* ortsgrupp*"~5

"grune* fpo*"~5 "grune* freiheitl*"~5 "grune* blau*"~5
"grune* spo"~5 "grune* sp"~5 "grune* sozialdemokrat*"~5
"grune* sozialist*"~5 "grune* rot*"~5
"grune* ovp"~5 "grune* vp"~5 "grune* volkspartei"~5 "grune* schwarz*"~5
"grune* neos"~5 "grune* kpo"~5 "grune* kommunist*"~5)

NOT("grune* (deutsch* cdu csu fdp jamaika spd afd ozdemir)"~7
"bundnis 90 grune*"~5 "grune* mark" )
```

[Precision: 0.87 | Recall: 0.89]

- NEOS (v9e)

```
neos "das neue osterreich"
```

[Precision: 0.90 | Recall: 0.95]

5.3 Lead Candidates

The appearance in news reports of the lead candidates in the 2024 national election was coded in v10. All variables have the value 1 when the given candidate is mentioned in the news report, and 0 if they are not mentioned.

- Herbert Kickl (v10a)
- Karl Nehammer (v10b)
- Andreas Babler (v10c)
- Werner Kogler (V10d)
- Beate Meindl-Reisinger (v10e)

5.4 Issues

We measure the appearance of issues by aggregating search string results on the finest level to the most general level that is reported in our data set. Issues were only validated on the most general level. Therefore, we recommend validating issue search strings on more fine-grained levels when needed. For all issues, the variable takes the value 1 if the issue is mentioned and 0 if it is not mentioned in the article.

- Economy and Budget (v11a): Aggregates the results of 49 search strings, covering the following topics: Economic policies, liberalization, deregulation, globalization, financial markets and banks, business policies, agriculture, tourism, state budget, government debt, and taxation policies.
- Labour Market and Employment (v11b): Aggregates the results of 39 search strings, covering the following topics: Labour market, unemployment, employee rights and welfare and consumer protection.
- Asylum and Immigration (v11c): Aggregates the results of 65 search strings, covering the following topics: Immigration, refugees, asylum, integration of immigrants, Islam, racism and anti-racism, citizenship policies, and the handling of immigrants concerning the welfare system, labour market, education, and crime.
- Family policy (v11d): Aggregates the results of 16 search strings, covering the following topics: Family support policies, childcare, parental leave, child welfare, gender equality in families, and work-family balance.

- Terrorism (v11e): Aggregates the results of 24 search strings, covering the following topics: Terrorist threats, counterterrorism policies, security measures, radicalization, and domestic and international terrorist activities.
- European integration (v11f): Aggregates the results of 75 search strings, covering the following topics: EU integration, EU treaties, EU extensions and exits, EU policies, EU participation and the Euro.
- Health and care (v11g): Aggregates the results of 36 search strings, covering the following topics: Healthcare systems, public health policies, hospitals, medical care, long-term care, and health insurance.
- Corruption (v11h): Aggregates the results of 9 search strings, covering the following topics: Political corruption, bribery, embezzlement, transparency, anti-corruption measures, and institutional integrity.
- Pensions (v11i): Aggregates the results of 19 search strings, covering the following topics: Pension schemes, retirement benefits, social security, elderly welfare, and policies for financial security of older populations.
- Tax policy (v11j): Aggregates the results of 56 search strings, covering the following topics: Taxation policies, income taxes, corporate taxes, tax reforms, tax incentives, and fiscal policy.
- Rent (v11k): Aggregates the results of 7 search strings, covering the following topics: Housing policies, rental regulations, real estate markets, public housing, urban planning, and affordability of living.
- Education and research (v11l): Aggregates the results of 58 search strings, covering the following topics: Educational policies, schools, universities, research, vocational education, adult education, and cultural policies.
- Climate change and environmental protection (v11m): Aggregates the results of 28 search strings, covering the following topics: Environmental protection, climate change and animal welfare.
- Justice (v11n): Aggregates the results of 9 search strings, covering the following topics: Judicial system, courts, legal reforms, criminal law, civil rights, and law enforcement.
- Media (v11o): Aggregates the results of 9 search strings, covering the following topics: Media coverage, journalism, press freedom, broadcasting, social media, and public communication.
- National defense (v11p): Aggregates the results of 19 search strings, covering the following topics: Military policy, defense strategies, armed forces, NATO and international defense cooperation, and homeland security.
- Inflation (v11q): Aggregates the results of 2 search strings, covering the following topics: Inflation trends, cost of living, price controls, economic impacts, and government responses to rising prices.

- Energy supply (v11r): Aggregates the results of 11 search strings, covering the following topics: Energy policies, electricity and gas supply, renewable energy, energy security, and sustainability of resources.
- COVID (v11s): Measured with one searchstring, covering different spellings of the term.

5.5 Sentiment

To calculate sentiment scores on the article level, we used two existing dictionaries for the German language: A sentiment dictionary for German political language by Christian Rauh[2] and the more general sentiment dictionary SentimentWortschatz/SentiWS[3]. Both dictionaries attribute positive and negative sentiments to certain cue words. The overall score of negativity or positivity was normalized by the number of words of the given paragraph. Note that numbers and punctation were removed from the texts as it is assumed that these do not bear any sentiment.

For both dictionaries, we provide a separate normalized score of positivity and negativity together with an overall normalized sentiment score. The normalized positivity score theoretically ranges from 0 (no positive sentiment cues) to 1 (all words of the text contain positive sentiment), whereas the normalized sentiment score ranges from 0 (no negative sentiment cues) to -1 (all words of the text contain negative sentiment). As the overall normalized sentiment score is calculated as the sum of the two separate sentiment scores, it theoretically ranges from -1 to 1. As both dictionaries have already been validated by their creators, we did not conduct additional validation of sentiment scores.

The following variables in the data set capture the sentiment of the news reports:

- Sentiment Rauh (v19a-c): Normalized sentiment scores calculated with the sentiment dictionary for German political language (Rauh, 2018). We provide information on sentiments in three ways: Positive sentiment – Rauh (v19a), Negative sentiment –Rauh (v19b) and Overall sentiment – Rauh (v19c).
- Sentiment SentiWS (v20a-c): Normalized sentiment scores calculated with the Sen-tiWS dictionary (Remus et al., 2010; v.2.0). We provide information on sentiments in three ways: Positive sentiment – SentiWS (v20a), Negative sentiment – SentiWS (v20b) and Overall sentiment – SentiWS (v20c).

[2]Rauh, C. (2018). *Replication Data for: Validating a sentiment dictionary for German political language (DRAFT VERSION)*. Harvard Dataverse. https://doi.org/10.7910/DVN/BKBXWD

[3]Remus, R., Quasthoff, U., & Heyer, G. (2010). *SentiWS-A Publicly Available German-language Resource for Sentiment Analysis*. LREC. http://www.lrec-conf.org/proceedings/lrec2010/pdf/490_Paper.pdf